

ULTRASOUND-GUIDED VACUUM-ASSISTED BIOPSY (US-VAB) IN A MULTIDISCIPLINARY PATHWAY FOR B3 PAPILLARY BREAST LESIONS: UPGRADE RATE AND CLINICAL OUTCOMES

PBK Tang¹, EPY Fung², WP Cheung², WS Mak², LW Lo², LKM Wong², DHY Cho²

¹ Department of Diagnostic and Interventional Radiology, Queen Elizabeth Hospital, Hong Kong

² Department of Diagnostic and Interventional Radiology, Kwong Wah Hospital, Hong Kong



BACKGROUND

B3 papillary breast lesions present diagnostic challenges necessitating a coordinated multidisciplinary approach to mitigate surgical overtreatment.

OBJECTIVES

To evaluate the outcomes of ultrasound-guided vacuum-assisted biopsy (US-VAB) for B3 papillary lesions as an alternative to diagnostic open surgical excision (SE).

Technical success was defined as immediate, complete sonographic removal of the lesion.

Long-term outcomes defined by recurrence or residual lesion on surveillance ultrasound.

METHODS

A retrospective cohort study evaluated consecutive CNB-proven B3 papillary lesions managed by therapeutic US-VAB between April 2022 and March 2024.

Patients lost to follow-up or those with concurrent ipsilateral same-quadrant breast malignancy as well as atypia were excluded.

RESULTS

Table 1: Histopathological Review Outcomes Following Therapeutic US-VAB

- Concordant Papillary (71.9%)
- Downgraded to Benign B2 (20.3%)
- Upgraded to Malignant (6.3%)

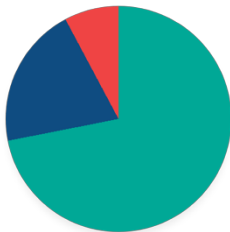
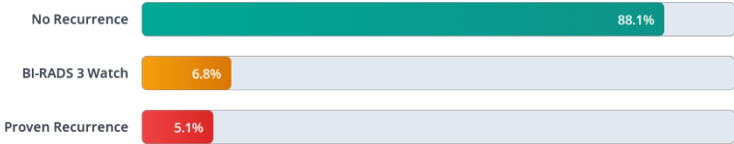


Table 2: Long-Term Surveillance Ultrasound Outcomes



KEY STATISTICAL OUTCOMES

92.2%

Primary Lesion Complete Removal

93.7%

Surgical Overtreatment Avoided

18.4

Months Mean Surveillance Follow-up

p < 0.001

Significance of Clearance vs. No Recurrence

92.2%

Immediate Complete Removal (59/64)

Zero

Major Procedural Complications

CONCLUSION / TAKE-HOME MESSAGE

US-VAB achieves complete immediate lesion removal in the vast majority of B3 papillary lesions and reliably identifies occult malignancies.

US-VAB yielded a high long-term recurrence-free rate when integrated within a multidisciplinary pathway.

This represents a paradigm shift away from diagnostic SE for non-atypical B3 papillary lesions, thereby optimizing both healthcare resources and patient outcomes.

REFERENCES

Park et al. (2020) evaluated upgrade rates in patients with benign papillary breast lesions, establishing an overall upgrade rate to malignancy of 10.6% and noting that cases without atypia can often be managed with radiological follow-up.
Girardi et al. (2021) analyzed a large cohort of B3 lesions undergoing VAB (including ultrasound-guided procedures) and confirmed that lesions without atypia demonstrate a significantly lower upgrade rate to malignancy, supporting active surveillance protocols.
Monib et al. (2021) assessed clinical outcomes and pathological upgrades following both stereotactic and ultrasound-guided vacuum-assisted breast biopsies for high-risk B3 lesions, tracking cases that were successfully downgraded to benign status.
Irmici et al. (2025) directly supports your abstract's "technical success" parameter by demonstrating that the complete radiological removal of B3 lesions during vacuum-assisted breast biopsy significantly lowers the subsequent upgrade rate to malignancy.

ACKNOWLEDGEMENTS

The authors declare no conflicts of interest.